Skooly Domain Model

This document describes **entities, relationships, and enums** for the Skooly school management platform. All diagrams match the current Prisma schema: **48 models** and **19 enums** across 7 domains. Diagrams are kept **separated by domain** for readability with cross-references documented at the end.

Implementation may still **fall back** to `Auth.schoolId` when no membership rows exist; see `docs/AUTH_SCHOOL_CONTEXT.md`.

Relationship Types (UML)

Symbol	Type	Meaning
`*--`	**Composition**	Strong ownership; child cannot exist without parent; lifecycle bound (required FK + cascade)
`o--`	**Aggregation**	Weak ownership; child can exist independently; optional FK or "has-a" collection
`-->`	**Association**	Uses/links to; no ownership; required or optional reference
`..>`	**Dependency**	Weak usage; one depends on another for operation (no FK)
`<|--`	**Generalization**	"Is-a" / specialization; child extends or depends on parent for identity (empty triangle arrow)

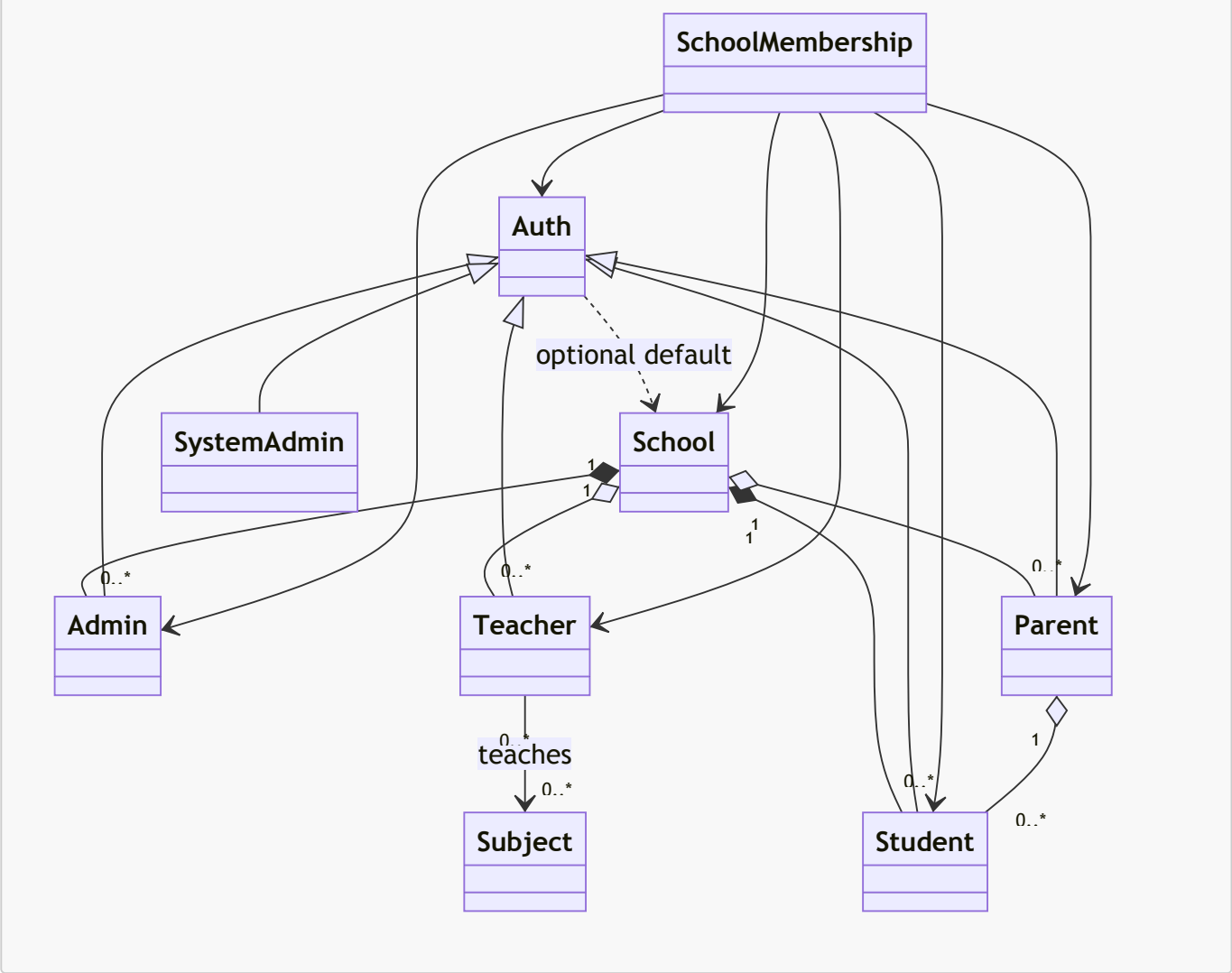
Diagram 1: Core Domain (People & Tenant)

Identity vs Tenant Access

Layer	Responsibility
Auth	**Identity** — who can sign in (session subject).
SchoolMembership	**Tenant access** — which School(s), role, and profile apply per context.

Rules:

- Prefer **active membership** for current school and role.
- **Teacher**, **Parent**, **School admin**: multiple active memberships (multi-school).
- **Student**: one active membership at a time.
- **System admin**: outside school membership graph.



Relationship descriptions:

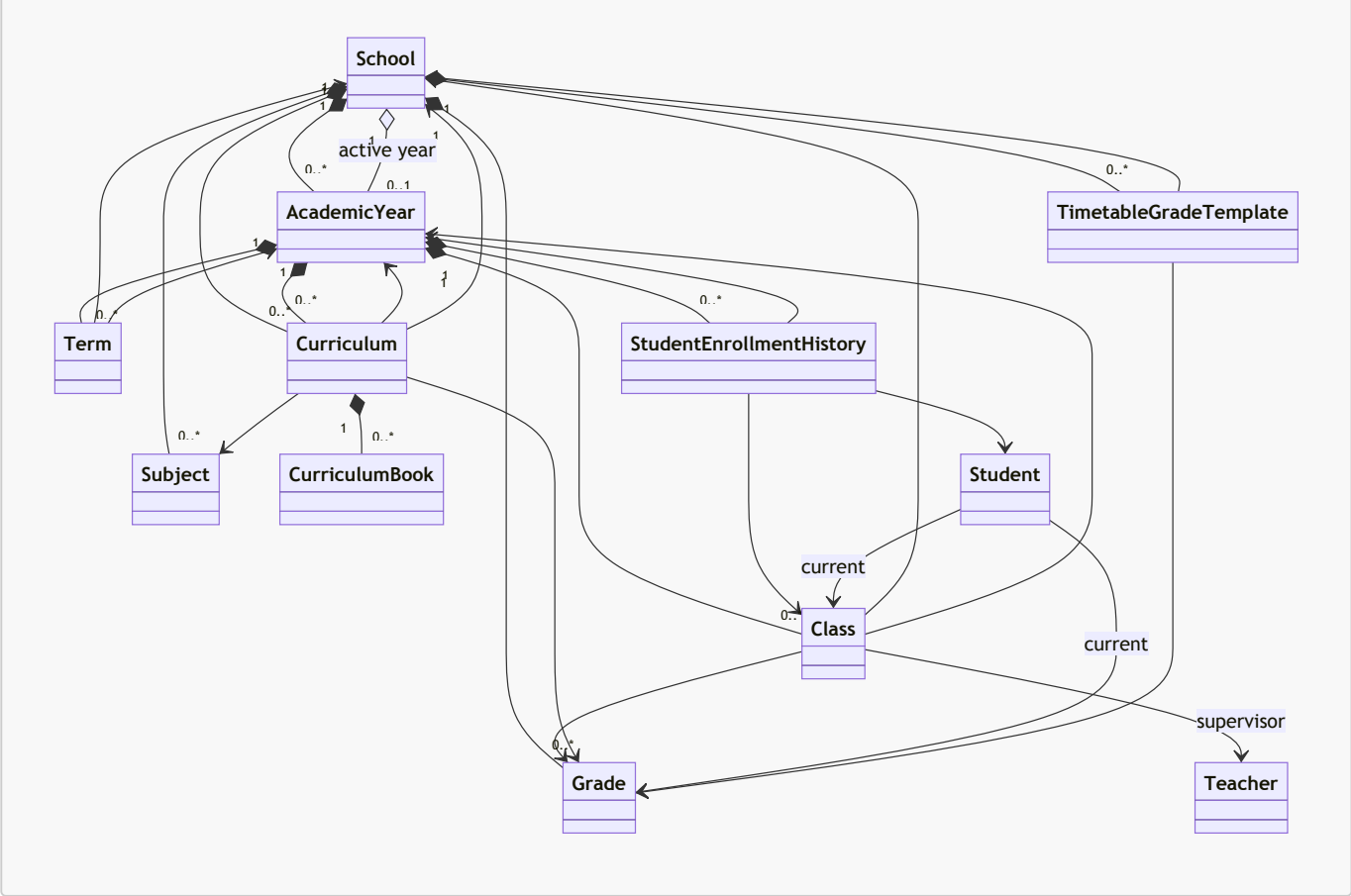
#	From	To	Type	Cardinality	Description
1	School	Admin	Composition	1 : 0..*	A school owns its admin profiles. Admin.schoolId is required; cascade delete.
2	School	Teacher	Aggregation	1 : 0..*	A school has teachers, but Teacher.schoolId is optional (multi-school teachers may have null home school).
3	School	Student	Composition	1 : 0..*	A school owns its students. Student.schoolId is required; cascade delete.
4	School	Parent	Aggregation	1 : 0..*	A school has parents, but Parent.schoolId is optional (multi-school parents).
5	Admin	Auth	Generalization	0..1 : 1	An admin is a specialization of Auth. authId is required and unique.

#	From	To	Type	Cardinality	Description
6	Teacher	Auth	Generalization	0..1 : 1	A teacher is a specialization of Auth. authId is required and unique.
7	Student	Auth	Generalization	0..1 : 1	A student is a specialization of Auth. authId is required and unique.
8	Parent	Auth	Generalization	0..1 : 1	A parent is a specialization of Auth. authId is required and unique.
9	SystemAdmin	Auth	Generalization	0..1 : 1	A system admin is a specialization of Auth. authId is required and unique.
10	Auth	School	Dependency	N : 0..1	Optional legacy default school on the auth row. Not authoritative for tenant; membership is.
11	SchoolMembership	Auth	Association	0..* : 1	Each membership references one Auth (the user). authId is required. Cascade delete.
12	SchoolMembership	School	Association	N : 1	Each membership row belongs to exactly one school. Required FK.
13	SchoolMembership	Admin	Association	N : 0..1	Optional pointer to the admin profile for this membership context.
14	SchoolMembership	Teacher	Association	N : 0..1	Optional pointer to the teacher profile for this membership context.
15	SchoolMembership	Student	Association	N : 0..1	Optional pointer to the student profile for this membership context.
16	SchoolMembership	Parent	Association	N : 0..1	Optional pointer to the parent profile for this membership context.
17	Parent	Student	Aggregation	1 : 0..*	A parent has zero or more children (students). Student.parentId is required.
18	Teacher	Subject	Association (M:N)	0..* : 0..*	Implicit many-to-many join table. A teacher can teach multiple subjects; a subject can have multiple teachers.

Membership cardinality:

Persona	Active rows per Auth
Teacher, Parent, School admin	0..* (multi-school)
Student	0..1 active
System admin	N/A

Diagram 2: Academic Structure



Relationship descriptions:

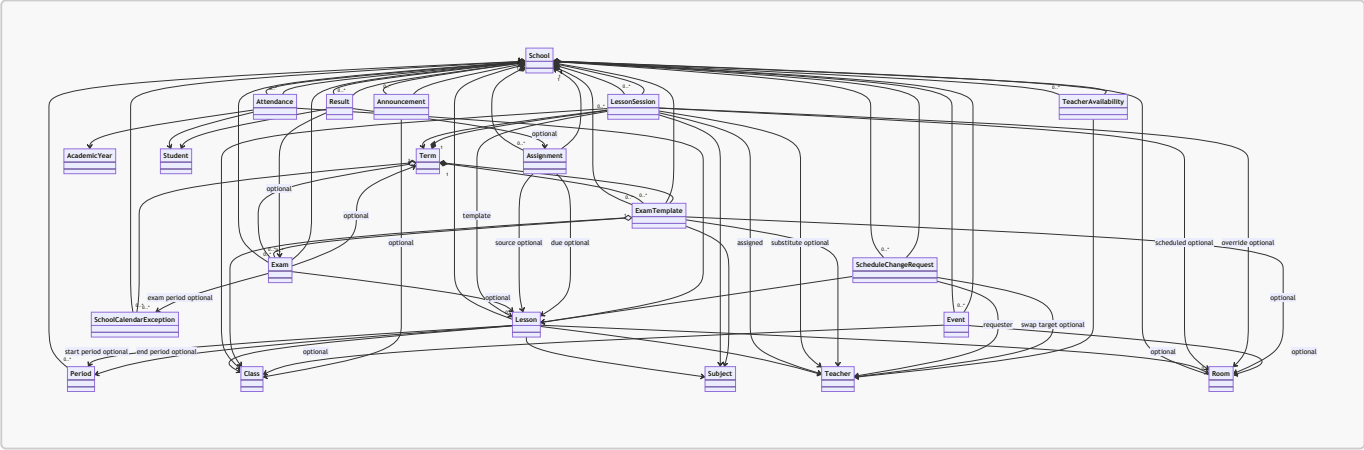
#	From	To	Type	Cardinality	Description
1	School	AcademicYear	Composition	1 : 0..*	A school owns its academic years. Required FK; cascade delete.
2	School	Grade	Composition	1 : 0..*	A school owns its grade levels (e.g. Grade 1-6). Required FK; cascade delete.
3	School	Subject	Composition	1 : 0..*	A school owns its subject catalog. Required FK; cascade delete.
4	School	Curriculum	Composition	1 : 0..*	A school owns curriculum rows (linking year x grade x subject). FK uses onDelete: NoAction.

#	From	To	Type	Cardinality	Description
5	School	TimetableGradeTemplate	Composition	1 : 0..*	A school owns timetable templates per grade. Required FK; cascade delete.
6	School	AcademicYear (active)	Aggregation	1 : 0..1	A school optionally points to one active academic year. SetNull on delete.
7	AcademicYear	Class	Composition	1 : 0..*	An academic year owns its classes. Required FK; cascade delete.
8	AcademicYear	Curriculum	Composition	1 : 0..*	An academic year owns its curriculum entries. Required FK; cascade delete.
9	AcademicYear	StudentEnrollmentHistory	Composition	1 : 0..*	An academic year owns enrollment history records. Required FK; cascade delete.
10	AcademicYear	Term	Composition	1 : 0..*	An academic year is divided into terms (e.g. Fall, Winter, Spring). Required FK; cascade delete.
11	Term	School	Association	N : 1	Each term belongs to one school. Required FK; cascade delete.

#	From	To	Type	Cardinality	Description
12	Term	AcademicYear	Association	N : 1	Each term belongs to one academic year. Required FK; cascade delete.
13	Class	School	Association	N : 1	Each class belongs to one school. Required FK; cascade delete.
14	Class	Grade	Association	N : 1	Each class is assigned to one grade level. Required FK.
15	Class	AcademicYear	Association	N : 1	Each class belongs to one academic year. Required FK; cascade delete.
16	Class	Teacher (supervisor)	Association	N : 0..1	A class may have one supervising teacher. Optional FK.
17	Curriculum	AcademicYear	Association	N : 1	Each curriculum entry belongs to one academic year.
18	Curriculum	Grade	Association	N : 1	Each curriculum entry targets one grade. Cascade delete.
19	Curriculum	Subject	Association	N : 1	Each curriculum entry is for one subject. Cascade delete.
20	Curriculum	School	Association	N : 1	Each curriculum entry belongs to one school (integrity guard).

#	From	To	Type	Cardinality	Description
21	Curriculum	CurriculumBook	Composition	1 : 0..*	A curriculum entry owns its book/resource records. Cascade delete.
22	StudentEnrollmentHistory	Student	Association	N : 1	Each enrollment record references one student. Cascade delete.
23	StudentEnrollmentHistory	Class	Association	N : 1	Each enrollment record references one class. Cascade delete.
24	StudentEnrollmentHistory	AcademicYear	Association	N : 1	Each enrollment record references one academic year. Cascade delete.
25	Student	Class	Association	N : 0..1	A student may be currently assigned to one class. Optional FK; SetNull on class delete.
26	Student	Grade	Association	N : 0..1	A student may be assigned a current grade level. Optional FK; SetNull on grade delete.
27	TimetableGradeTemplate	School	Association	N : 1	Each template belongs to one school. Cascade delete. Unique per (school, grade).
28	TimetableGradeTemplate	Grade	Association	N : 1	Each template targets one grade. Cascade delete.

Diagram 3: Scheduling, Calendar & Academic Activities



Relationship descriptions:

#	From	To	Type	Cardinality	Description
1	School	Period	Composition	1 : 0..*	A school owns its bell schedule periods. Cascade delete.
2	School	Lesson	Composition	1 : 0..*	A school owns weekly lesson templates. Cascade delete.
3	School	LessonSession	Composition	1 : 0..*	A school owns dated lesson instances. Cascade delete.
4	School	ExamTemplate	Composition	1 : 0..*	A school owns recurring exam templates. Cascade delete.
5	School	Exam	Composition	1 : 0..*	A school owns exam records. Cascade delete.
6	School	Assignment	Composition	1 : 0..*	A school owns assignment records. Cascade delete.
7	School	Attendance	Composition	1 : 0..*	A school owns attendance records. Cascade delete.

#	From	To	Type	Cardinality	Description
8	School	Result	Composition	1 : 0..*	A school owns result/score records. Cascade delete.
9	School	Room	Composition	1 : 0..*	A school owns its physical rooms. Cascade delete.
10	School	Event	Composition	1 : 0..*	A school owns calendar events. Cascade delete.
11	School	Announcement	Composition	1 : 0..*	A school owns announcements. Cascade delete.
12	School	TeacherAvailability	Composition	1 : 0..*	A school owns teacher availability slots. Cascade delete.
13	School	ScheduleChangeRequest	Composition	1 : 0..*	A school owns schedule change requests. Cascade delete.
14	School	SchoolCalendarException	Composition	1 : 0..*	A school owns calendar exceptions (holidays, breaks, exam periods). Cascade delete.
15	Term	LessonSession	Composition	1 : 0..*	A term owns its generated lesson sessions. Cascade delete.
16	Term	ExamTemplate	Composition	1 : 0..*	A term owns recurring exam templates.
17	Term	SchoolCalendarException	Composition	1 : 0..*	A term owns its calendar exceptions. Cascade delete.

#	From	To	Type	Cardinality	Description
18	Term	Exam	Aggregation	1 : 0..*	A term may have exams linked to it. Exam.termId is optional; SetNull on delete.
19	Lesson	Class	Association	N : 1	Each lesson template is for one class. Required FK.
20	Lesson	Subject	Association	N : 1	Each lesson template teaches one subject. Required FK.
21	Lesson	Teacher	Association	N : 1	Each lesson template is assigned to one teacher. Required FK.
22	Lesson	Room	Association	N : 0..1	A lesson may be assigned a room. Null for online lessons. SetNull on delete.
23	Lesson	Period (start)	Association	N : 0..1	A lesson may map to a start bell period. SetNull on delete.
24	Lesson	Period (end)	Association	N : 0..1	For multi-block lessons, an end period. Null means single period. SetNull on delete.
25	LessonSession	Term	Association	N : 1	Each session belongs to one term. Cascade delete.
26	LessonSession	Lesson (template)	Association	N : 1	Each session is generated from one lesson template. Cascade delete.

#	From	To	Type	Cardinality	Description
27	LessonSession	School	Association	N : 1	Each session belongs to one school. Cascade delete.
28	LessonSession	Class	Association	N : 1	Each session is for one class. Required FK.
29	LessonSession	Subject	Association	N : 1	Each session is for one subject (denormalized from template). Required FK.
30	LessonSession	Teacher (assigned)	Association	N : 1	Primary teacher for this session. Required FK.
31	LessonSession	Room (scheduled)	Association	N : 0..1	Scheduled room (from template). Optional; SetNull on delete.
32	LessonSession	Teacher (substitute)	Association	N : 0..1	Optional substitute teacher override. SetNull on delete.
33	LessonSession	Room (override)	Association	N : 0..1	Optional room override (admin moved the session). SetNull on delete.
34	ExamTemplate	School	Association	N : 1	Each exam template belongs to one school. Cascade delete.
35	ExamTemplate	Term	Association	N : 1	Each exam template is scoped to one term.
36	ExamTemplate	Class	Association	N : 1	Each exam template targets one class.

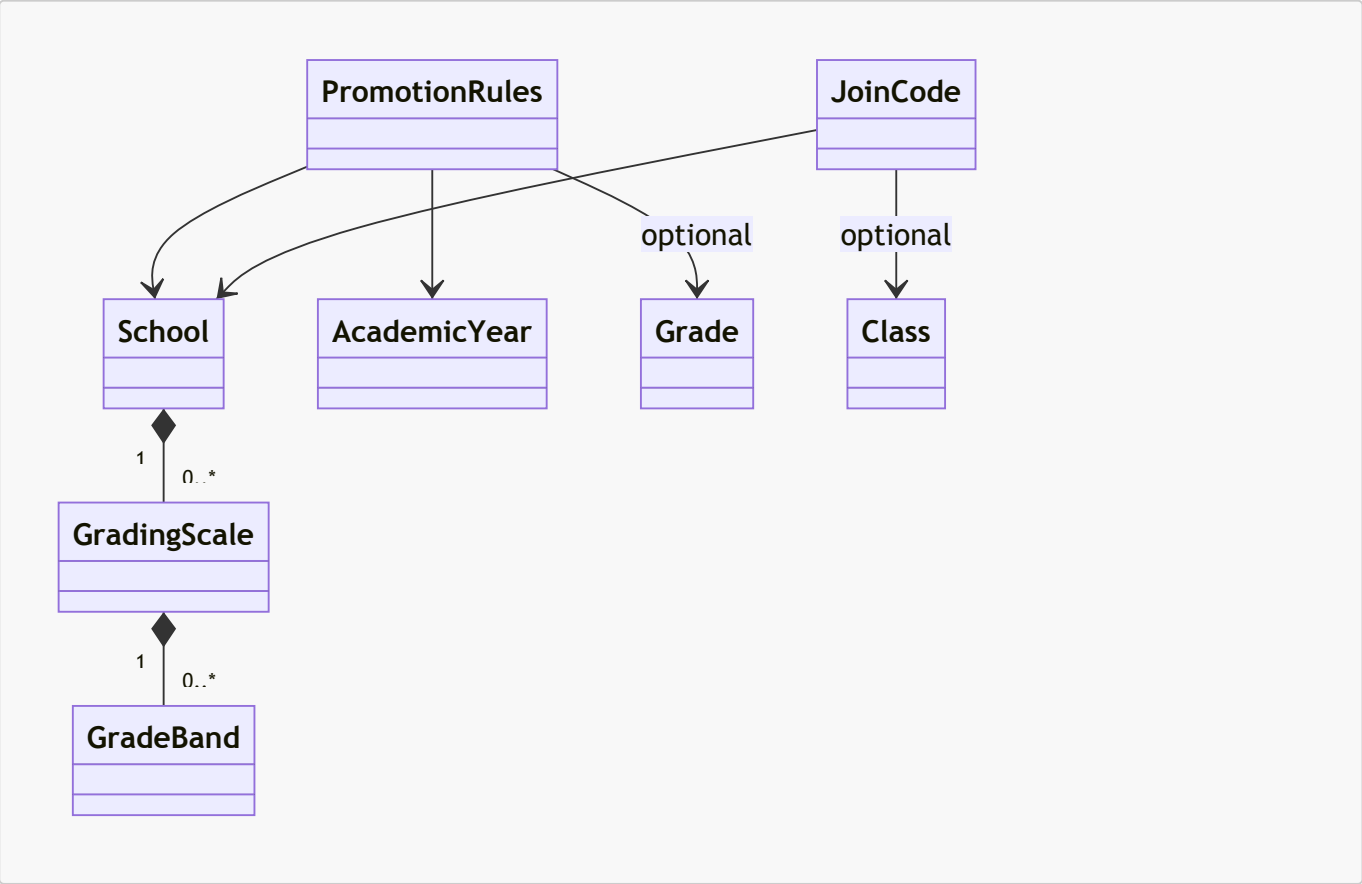
#	From	To	Type	Cardinality	Description
37	ExamTemplate	Subject	Association	N : 1	Each exam template is for one subject.
38	ExamTemplate	Teacher	Association	N : 1	Each exam template is assigned to one teacher.
39	ExamTemplate	Room	Association	N : 0..1	Optional room for the exam.
40	ExamTemplate	Exam	Aggregation	1 : 0..*	A template generates zero or more exam instances. SetNull on delete.
41	Exam	School	Association	N : 1	Each exam belongs to one school. Cascade delete.
42	Exam	Lesson	Association	N : 0..1	An exam may link to a lesson (subject/class context). Optional FK.
43	Exam	Term	Association	N : 0..1	An exam may belong to a term. Optional FK; SetNull on delete.
44	Exam	SchoolCalendarException	Association	N : 0..1	An exam may fall within a declared exam period window. Optional FK; SetNull on delete.
45	Assignment	School	Association	N : 1	Each assignment belongs to one school. Cascade delete.

#	From	To	Type	Cardinality	Description
46	Assignment	Lesson (source)	Association	N : 0..1	The lesson that assigned this work. Optional FK; SetNull on delete.
47	Assignment	Lesson (due)	Association	N : 0..1	The lesson on which this assignment is due. Optional FK; SetNull on delete.
48	Attendance	School	Association	N : 1	Each attendance record belongs to one school. Cascade delete.
49	Attendance	Lesson	Association	N : 1	Each attendance record is for one lesson. Required FK.
50	Attendance	Student	Association	N : 1	Each attendance record is for one student. Required FK.
51	Attendance	AcademicYear	Association	N : 1	Each attendance record belongs to one academic year.
52	Result	School	Association	N : 1	Each result belongs to one school. Cascade delete.
53	Result	Student	Association	N : 1	Each result is for one student. Required FK.
54	Result	Exam	Association	N : 0..1	A result may be for an exam. Optional FK (XOR with assignment at app level).

#	From	To	Type	Cardinality	Description
55	Result	Assignment	Association	N : 0..1	A result may be for an assignment. Optional FK (XOR with exam at app level).
56	ScheduleChangeRequest	School	Association	N : 1	Each request belongs to one school. Cascade delete.
57	ScheduleChangeRequest	Lesson	Association	N : 1	Each request targets one lesson. Cascade delete.
58	ScheduleChangeRequest	Teacher (requester)	Association	N : 1	The teacher who submitted the request. Cascade delete.
59	ScheduleChangeRequest	Teacher (swap target)	Association	N : 0..1	For SWAP type: the proposed swap teacher. Optional; NoAction on delete.
60	TeacherAvailability	Teacher	Association	N : 1	Each availability slot belongs to one teacher. Cascade delete.
61	TeacherAvailability	School	Association	N : 1	Each availability slot is scoped to one school. Cascade delete. Unique per (teacher, day, time, school).
62	Event	School	Association	N : 1	Each event belongs to one school. Cascade delete.

#	From	To	Type	Cardinality	Description
63	Event	Class	Association	N : 0..1	An event may target a specific class. Optional FK; cascade delete.
64	Event	Room	Association	N : 0..1	An event may be in a room. Optional FK; SetNull on delete.
65	Announcement	School	Association	N : 1	Each announcement belongs to one school. Cascade delete.
66	Announcement	Class	Association	N : 0..1	An announcement may target a specific class. Optional FK.

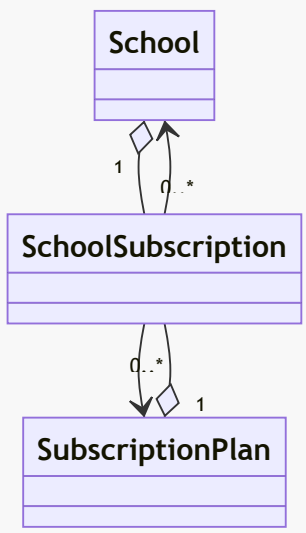
Diagram 4: Grading, Promotion & Join Access



Relationship descriptions:

#	From	To	Type	Cardinality	Description
1	School	GradingScale	Composition	1 : 0..*	A school owns its grading scales. Cascade delete.
2	GradingScale	GradeBand	Composition	1 : 0..*	A grading scale owns its band entries (letter grades / ranges). Cascade delete.
3	PromotionRules	School	Association	N : 1	Each promotion rule set belongs to one school. Cascade delete.
4	PromotionRules	AcademicYear	Association	N : 1	Each promotion rule set is for one academic year.
5	PromotionRules	Grade	Association	N : 0..1	A promotion rule set may target a specific grade. Optional FK. Unique per (school, year, grade).
6	JoinCode	School	Association	N : 1	Each join code belongs to one school. Cascade delete.
7	JoinCode	Class	Association	N : 0..1	A join code may be scoped to one class (for CLASS_STUDENT type). Optional FK; cascade delete.

Diagram 5: Billing



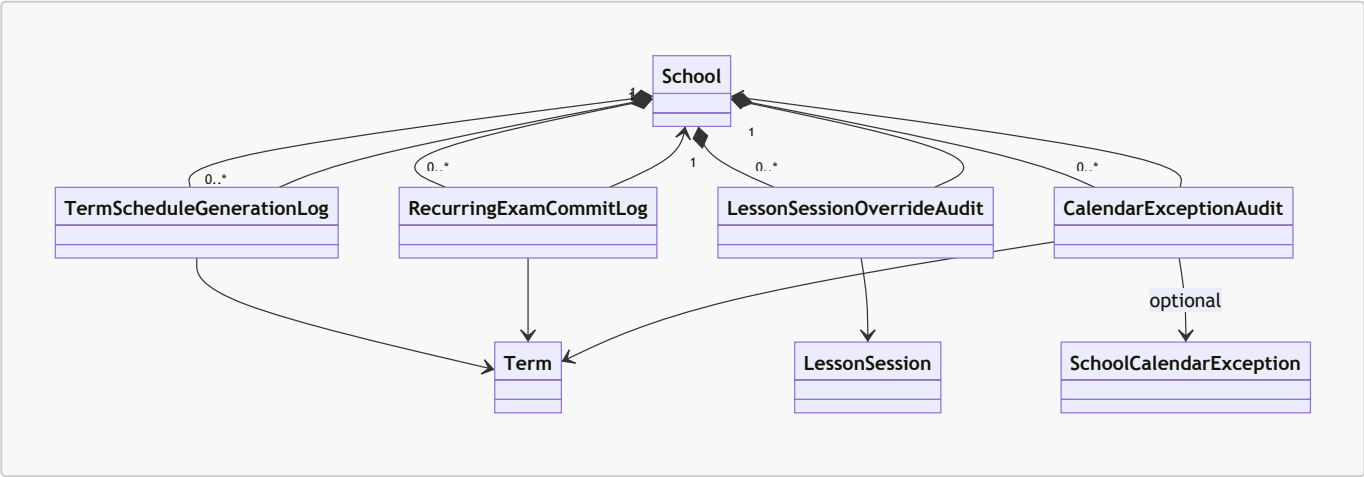
Relationship descriptions:

#	From	To	Type	Cardinality	Description
1	School	SchoolSubscription	Aggregation	1 : 0..*	A school can have multiple subscriptions (active, past, etc.).

#	From	To	Type	Cardinality	Description
2	SubscriptionPlan	SchoolSubscription	Aggregation	1 : 0..*	A plan can be subscribed to by many schools.
3	SchoolSubscription	School	Association	N : 1	Each subscription belongs to one school. Required FK.
4	SchoolSubscription	SubscriptionPlan	Association	N : 1	Each subscription references one plan. Required FK.

Diagram 6: Scheduling & Calendar Operations (Audit)

Audit tables record pipeline runs and administrative edits. They support troubleshooting and accountability.



Relationship descriptions:

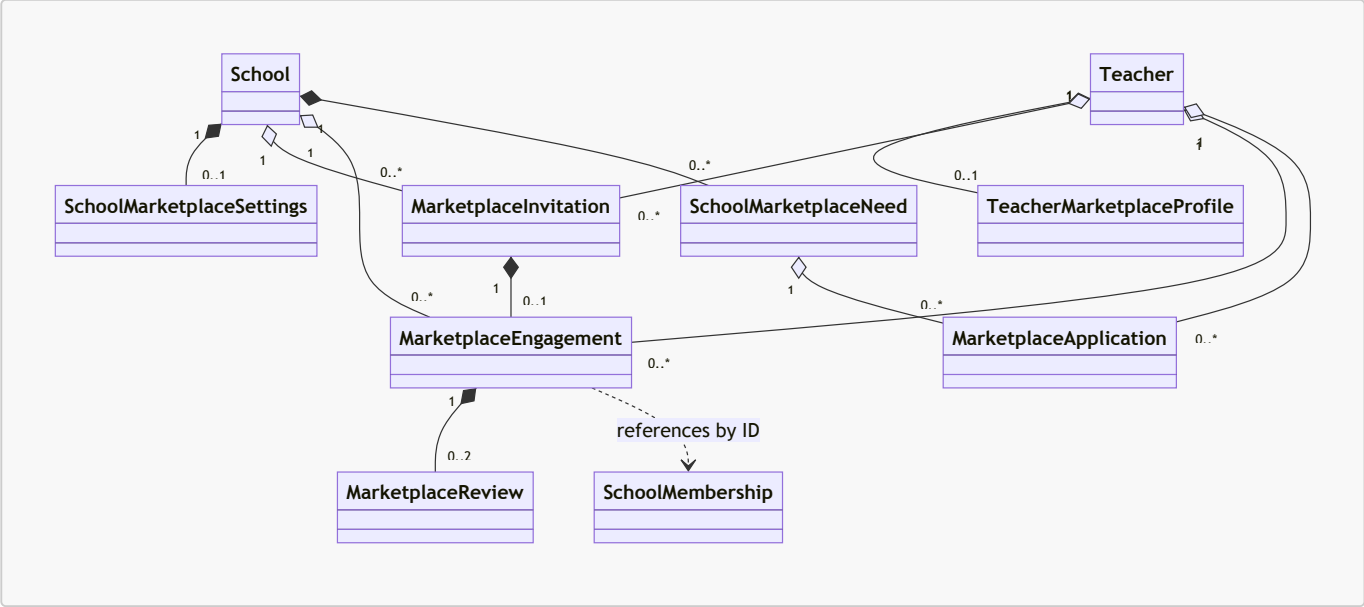
#	From	To	Type	Cardinality	Description
1	School	TermScheduleGenerationLog	Composition	1 : 0..*	A school owns its generation run logs. Cascade delete.
2	School	RecurringExamCommitLog	Composition	1 : 0..*	A school owns its recurring exam commit logs. Cascade delete.

#	From	To	Type	Cardinality	Description
3	School	LessonSessionOverrideAudit	Composition	1 : 0..*	A school owns lesson session override audits. Cascade delete.
4	School	CalendarExceptionAudit	Composition	1 : 0..*	A school owns calendar exception audits. Cascade delete.
5	TermScheduleGenerationLog	Term	Association	N : 1	Each log references the term it was generated for. Cascade delete.
6	RecurringExamCommitLog	Term	Association	N : 1	Each log references the term. Cascade delete.
7	LessonSessionOverrideAudit	LessonSession	Association	N : 1	Each audit references the session that was overridden. Cascade delete.
8	CalendarExceptionAudit	Term	Association	N : 1	Each audit references the term. Cascade delete.

#	From	To	Type	Cardinality	Description
9	CalendarExceptionAudit	SchoolCalendarException	Association	N : 0..1	May reference the exception (null if the exception was deleted). SetNull on delete.

Diagram 7: Teacher Marketplace

The marketplace enables two-sided discovery: teachers opt in with a profile; schools opt in via settings; invitations flow from schools to teachers; accepted invitations create SchoolMemberships and tracked engagements; reviews build trust; needs boards let schools post openings and teachers apply.



Relationship descriptions:

#	From	To	Type	Cardinality	Description
1	Teacher	TeacherMarketplaceProfile	Composition	1 : 0..1	A teacher may opt in with one marketplace profile. Cascade delete.
2	School	SchoolMarketplaceSettings	Composition	1 : 0..1	A school has at most one marketplace settings row. Cascade delete.

#	From	To	Type	Cardinality	Description
3	School	MarketplaceInvitation	Aggregation	1 : 0..*	A school can send many invitations. Cascade delete.
4	Teacher	MarketplaceInvitation	Aggregation	1 : 0..*	A teacher can receive many invitations from different schools. Cascade delete.
5	MarketplaceInvitation	MarketplaceEngagement	Composition	1 : 0..1	An accepted invitation creates exactly one engagement. Unique FK on invitationId.
6	School	MarketplaceEngagement	Aggregation	1 : 0..*	A school can have many active engagements. Cascade delete.
7	Teacher	MarketplaceEngagement	Aggregation	1 : 0..*	A teacher can have engagements at multiple schools. Cascade delete.
8	MarketplaceEngagement	MarketplaceReview	Composition	1 : 0..2	Each engagement can have at most two reviews: one from the school, one from the teacher. Unique per (engagement, reviewerRole). Cascade delete.
9	MarketplaceEngagement	SchoolMembership	Dependency	N : 0..1	References a SchoolMembership by ID (no FK constraint). Created on acceptance to integrate teacher into school's scheduling system.
10	School	SchoolMarketplaceNeed	Composition	1 : 0..*	A school can post many open positions. Cascade delete.

#	From	To	Type	Cardinality	Description
11	SchoolMarketplaceNeed	MarketplaceApplication	Aggregation	1 : 0..*	A posted need can receive many applications. Cascade delete.
12	Teacher	MarketplaceApplication	Aggregation	1 : 0..*	A teacher can apply to many needs. Unique per (need, teacher). Cascade delete.

Marketplace flow:

1. Teacher publishes a TeacherMarketplaceProfile (subject tags, availability, rate).
2. School enables marketplace via SchoolMarketplaceSettings.
3. Admin searches profiles (excludes own school teachers) and sends MarketplaceInvitation.
4. Teacher accepts -> transaction creates SchoolMembership + MarketplaceEngagement.
5. Teacher is now in the school's teaching pool (visible to timetable assistant, assignable to lessons).
6. Admin ends engagement -> optionally deactivates membership -> both sides can leave MarketplaceReview.
7. Schools post SchoolMarketplaceNeed; teachers with matching subjects see them and submit MarketplaceApplication.

Enums Reference

19 enums define the valid values for status fields, types, and categories across the domain.

Diagram 1 (Core)

Enum	Values	Used By
AccountType	`SCHOOL_ADMIN`, `TEACHER`, `STUDENT`, `PARENT`, `SYSTEM_ADMIN`	Auth.accountType — determines the type of account at creation.
UserSex	`MALE`, `FEMALE`, `OTHER`	Teacher.sex, Student.sex — demographic field on profiles.

Diagram 2 (Academic)

Enum	Values	Used By
Day	`MONDAY` .. `SUNDAY`	Lesson.day, LessonSession.day, TeacherAvailability.dayOfWeek — weekday for scheduling.
EnrollmentStatus	`ENROLLED`, `PROMOTED`, `REPEATED`, `WITHDRAWN`, `COMPLETED`, `GRADUATED`	StudentEnrollmentHistory.status — lifecycle of a student's enrollment in a class.
CurriculumBookRole	`primary`, `supplementary`, `workbook`, `reader`, `teacher`, `digital`, `other`	CurriculumBook.role — classification of a teaching resource.

Diagram 3 (Scheduling & Activities)

Enum	Values	Used By
LessonDeliveryMode	`IN_PERSON`, `ONLINE`	Lesson.deliveryMode, LessonSession.deliveryMode — whether teaching is physical or remote. Both count toward teacher overlap.
LessonSessionStatus	`SCHEDULED`, `CANCELLED`	LessonSession.status — runtime state of a generated session instance.
CalendarExceptionType	`HOLIDAY`, `BREAK`, `EXAM_PERIOD`	SchoolCalendarException.type — what kind of calendar block this is. Generation skips days inside any type.
ExamCategory	`COURSE_EXAM`, `POP_QUIZ`	Exam.examCategory — distinguishes full exams from in-class quizzes (affects calendar badges).
AttendanceStatus	`PRESENT`, `ABSENT`, `LATE`	Attendance.status — per-student per-lesson attendance mark.
ScheduleChangeType	`TIME_CHANGE`, `SWAP`	ScheduleChangeRequest.requestedChangeType — what kind of schedule change a teacher requests.
RequestStatus	`PENDING`, `APPROVED`, `REJECTED`, `CANCELED`	ScheduleChangeRequest.status — lifecycle of a teacher's schedule change request.

Diagram 4 (Grading & Join)

Enum	Values	Used By
JoinCodeType	`CLASS_STUDENT`, `TEACHER_INVITE`, `PARENT_LINK`	JoinCode.type — determines the onboarding flow: enroll parent+student, invite a teacher, or link a parent to an existing student.

Diagram 5 (Billing)

Enum	Values	Used By
BillingCycle	`MONTHLY`, `YEARLY`	SubscriptionPlan.billingCycle — payment frequency.
SubscriptionStatus	`ACTIVE`, `CANCELED`, `PAST_DUE`, `INCOMPLETE`, `INCOMPLETE_EXPIRED`, `TRIALING`, `UNPAID`	SchoolSubscription.status — Stripe subscription lifecycle states.

Diagram 7 (Marketplace)

Enum	Values	Used By
MarketplaceInvitationStatus	`PENDING`, `ACCEPTED`, `DECLINED`, `WITHDRAWN`, `EXPIRED`	MarketplaceInvitation.status — lifecycle of a school-to-teacher invitation.

Enum	Values	Used By
EngagementStatus	`ACTIVE`, `COMPLETED`, `TERMINATED`	MarketplaceEngagement.status — lifecycle of a working relationship.
ReviewerRole	`SCHOOL`, `TEACHER`	MarketplaceReview.reviewerRole — which side left the review. Unique per (engagement, role).
ApplicationStatus	`PENDING`, `REVIEWED`, `ACCEPTED`, `REJECTED`	MarketplaceApplication.status — lifecycle of a teacher's application to an open position.

Cross-Diagram Dependencies

Overview

```
Diagram 1 (Core) --> Diagram 2 (Academic): School, Teacher, Student
Diagram 2 (Academic) --> Diagram 3 (Scheduling): Class, Subject, Teacher,
Student, Term, AcademicYear
Diagram 1 (Core) --> Diagram 4 (Grading / Join): School, Grade, AcademicYear
Diagram 1 (Core) --> Diagram 5 (Billing): School
Diagrams 2 & 3 --> Diagram 6 (Audit): School, Term, LessonSession,
SchoolCalendarException
Diagram 1 (Core) --> Diagram 7 (Marketplace): Teacher, School, SchoolMembership
```

Dependency Flow

```
                    +-------------+
                    |  Diagram 1  |
                    |   (Core)    |
                    +------+------+
                           |
        +-----------------+-----------------+-----------+
        |                 |                 |           |
        v                 v                 v           v
  +------------+    +------------+    +------------+ +------------+
  | Diagram 2  |    | Diagram 4  |    | Diagram 5  | | Diagram 7  |
  | (Academic) |    | (Grade/    |    | (Billing)  | |(Marketplace)|
  |            |    |  Join)     |    |            | |            |
  +-----+------+    +------------+    +------------+ +-----+------+
        |                                                  |
        v                                                  |
  +------------+    +------------+                         |
  | Diagram 3  |--->| Diagram 6  |                         |
  | (Schedule) |    | (Audit)    |                         |
  +------+-----+    +------------+                         |
         |                                                  |
         +------------- SchoolMmembership ------------------+
```

Key cross-diagram links:

- **Marketplace -> Core -> Scheduling**: Accepted invitations create SchoolMembership rows (Diagram 1), making marketplace teachers available to the scheduling system (Diagram 3).
- **Scheduling -> Academic**: Lessons, sessions, exams, and attendance all reference Class, Subject, Teacher, and Term from the academic structure.
- **Audit -> Scheduling**: Audit tables reference Term, LessonSession, and SchoolCalendarException.

Typical Setup Order

School and people -> academic year, grades, subjects, terms -> classes, curriculum, enrollments -> bell periods and weekly lessons -> generated lesson sessions and exams -> assignments, attendance, results. Billing, grading, and marketplace configuration can run in parallel once the school exists.

Reverse Dependencies (Queries)

- **Student schedule (dated):** Student -> Class -> LessonSession (by Term).
- **Teacher schedule:** Teacher -> Lesson and LessonSession by date; merged across schools via SchoolMembership.
- **Parent dashboard:** Parent -> Students (across all SchoolMembership schools) -> per-child schedule, grades, attendance.
- **Exam calendar:** Exam -> Term, optional ExamTemplate, optional SchoolCalendarException.
- **Marketplace search:** TeacherMarketplaceProfile (published, excluding own school) -> Teacher -> reviews via MarketplaceEngagement.
- **Subscription limits:** School -> SchoolSubscription -> SubscriptionPlan caps.

Entity Count Summary

Diagram	Models	Count
1. Core	School, Auth, Admin, Teacher, Student, Parent, SystemAdmin, SchoolMembership	8
2. Academic	AcademicYear, Term, Grade, Class, Subject, Curriculum, CurriculumBook, StudentEnrollmentHistory, TimetableGradeTemplate	9
3. Scheduling	Period, Lesson, LessonSession, ExamTemplate, Exam, Assignment, Attendance, Result, Room, ScheduleChangeRequest, TeacherAvailability, Event, Announcement, SchoolCalendarException	14
4. Grading/Join	GradingScale, GradeBand, PromotionRules, JoinCode	4
5. Billing	SubscriptionPlan, SchoolSubscription	2
6. Audit	TermScheduleGenerationLog, RecurringExamCommitLog, LessonSessionOverrideAudit, CalendarExceptionAudit	4
7. Marketplace	TeacherMarketplaceProfile, SchoolMarketplaceSettings, MarketplaceInvitation, MarketplaceEngagement, MarketplaceReview, SchoolMarketplaceNeed, MarketplaceApplication	7
Total		**48**